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Set-9

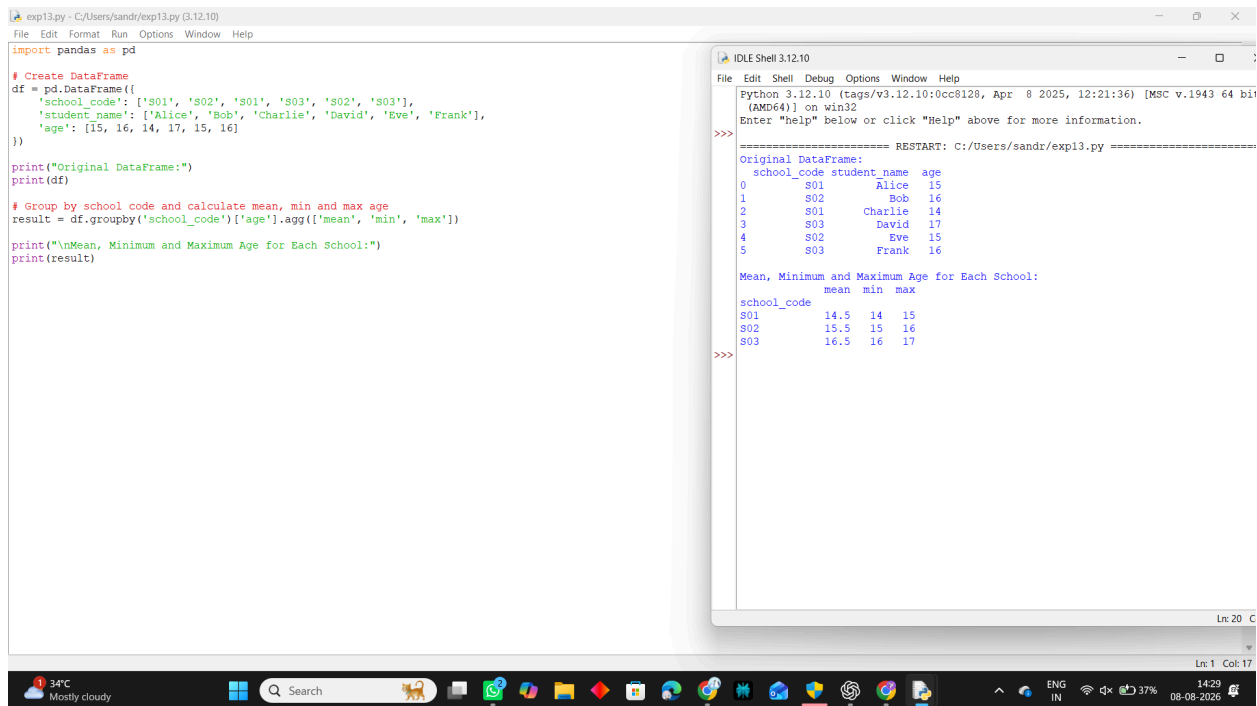
17. Write a Pandas program to split the following dataframe by school code and get mean, min, and max value of age for each school.

Aim

To write a Pandas program to split a DataFrame based on **school code** and find the **mean, minimum, and maximum age** of students in each school.

Procedure

1. Import the Pandas library.
2. Create a DataFrame containing **school_code** and **age**.
3. Group the DataFrame using **groupby()** based on **school_code**.
4. Apply **mean()**, **min()**, and **max()** functions to the **age** column.
5. Display the results for each school.



The screenshot shows a Python IDE with two windows. The left window displays the source code for a Pandas program, and the right window shows the output of the program.

```
exp13.py - C:/Users/sandr/exp13.py (3.12.10)
File Edit Format Run Options Window Help
import pandas as pd

# Create DataFrame
df = pd.DataFrame({
    'school_code': ['S01', 'S02', 'S01', 'S03', 'S02', 'S03'],
    'student_name': ['Alice', 'Bob', 'Charlie', 'David', 'Eve', 'Frank'],
    'age': [15, 16, 14, 17, 15, 16]
})

print("Original DataFrame:")
print(df)

# Group by school code and calculate mean, min and max age
result = df.groupby('school_code')['age'].agg(['mean', 'min', 'max'])

print("\nMean, Minimum and Maximum Age for Each School:")
print(result)
```

The output window shows the following results:

```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit
(AMD64)] on win32
Enter "help" below or click "Help" above for more information.

===== RESTART: C:/Users/sandr/exp13.py =====

Original DataFrame:
  school_code student_name  age
0         S01         Alice   15
1         S02          Bob   16
2         S01        Charlie   14
3         S03         David   17
4         S02          Eve   15
5         S03         Frank   16

Mean, Minimum and Maximum Age for Each School:
              mean  min  max
school_code
S01             14.5   14   15
S02             15.5   15   16
S03             16.5   16   17
```

Result

Thus, the DataFrame was successfully split based on **school code**, and the **mean, minimum, and maximum age** were calculated for each school.

18. Write a Pandas program to split the following given dataframe into groups based on

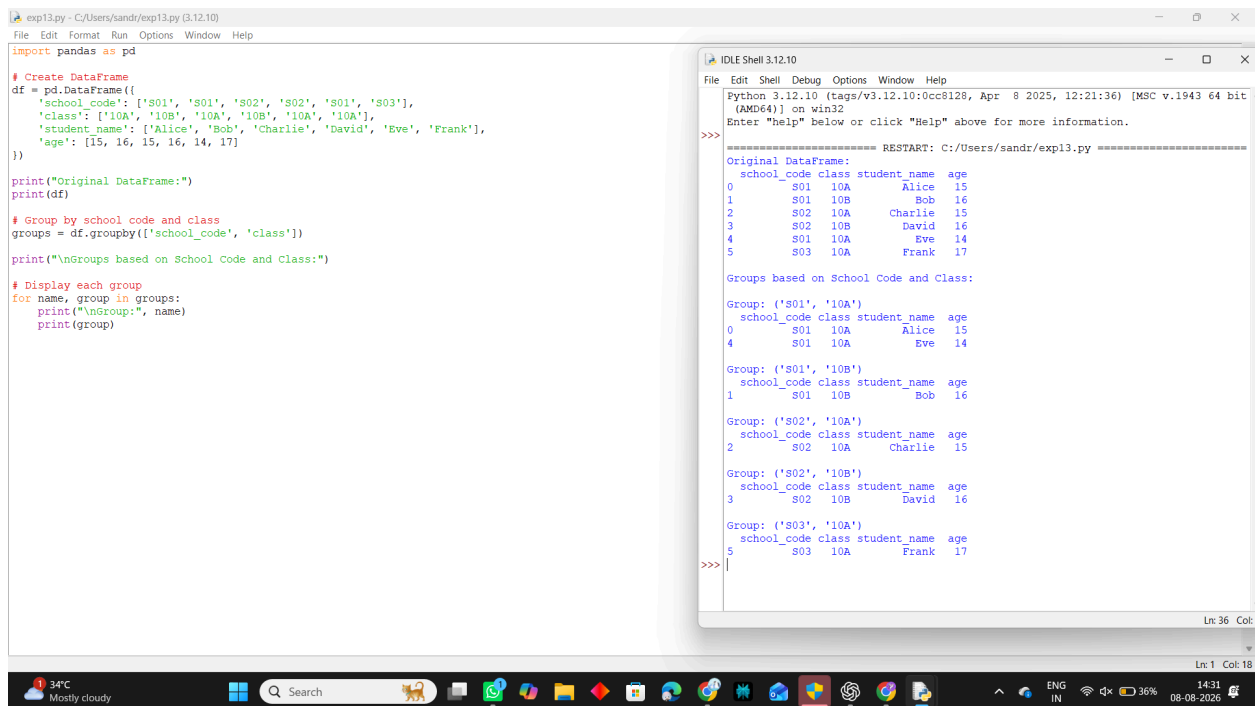
school code and class.

Aim

To write a Pandas program to split a given DataFrame into groups based on **school code** and **class**.

Procedure

1. Import the Pandas library.
2. Create a DataFrame containing **school_code**, **class**, **student_name**, and **age**.
3. Use the **groupby()** function with **school_code** and **class**.
4. Display each group separately.
5. Verify that the records are divided according to both school code and class.



```
exp13.py - C:/Users/sandr/exp13.py (3.12.10)
File Edit Format Run Options Window Help

import pandas as pd

# Create DataFrame
df = pd.DataFrame({
    'school_code': ['S01', 'S01', 'S02', 'S02', 'S01', 'S03'],
    'class': ['10A', '10B', '10A', '10B', '10A', '10A'],
    'student_name': ['Alice', 'Bob', 'Charlie', 'David', 'Eve', 'Frank'],
    'age': [15, 16, 15, 16, 14, 17]
})

print("Original DataFrame:")
print(df)

# Group by school code and class
groups = df.groupby(['school_code', 'class'])

print("\nGroups based on School Code and Class:")

# Display each group
for name, group in groups:
    print("\nGroup:", name)
    print(group)
```

```
IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help

Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit
(AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/sandr/exp13.py =====

Original DataFrame:
  school_code class student_name age
0         S01  10A         Alice  15
1         S01  10B          Bob  16
2         S02  10A        Charlie  15
3         S02  10B          David  16
4         S01  10A          Eve  14
5         S03  10A          Frank  17

Groups based on School Code and Class:

Group: ('S01', '10A')
  school_code class student_name age
0         S01  10A         Alice  15
4         S01  10A          Eve  14

Group: ('S01', '10B')
  school_code class student_name age
1         S01  10B          Bob  16

Group: ('S02', '10A')
  school_code class student_name age
2         S02  10A        Charlie  15

Group: ('S02', '10B')
  school_code class student_name age
3         S02  10B          David  16

Group: ('S03', '10A')
  school_code class student_name age
5         S03  10A          Frank  17

>>>
```

Result

Thus, the given DataFrame was successfully **split into groups based on both school code and class** using the Pandas **groupby()** function.